WASIM AKHTAR

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EDUCATION

Presidency University, Bangalore *Sep 2023 – Jun/Jul 2027 (Expected)*

B.Tech. Computer Science & Technology (Artificial Intelligence & Machine Learning) | CGPA: 6.73/10 (through Semester 6)

Relevant Coursework: Artificial Intelligence & Machine Learning · Deep Learning & Reinforcement Learning · Explainable AI · Edge AI & IoT Analytics · Operating Systems · Data Structures & Algorithms · Database Management Systems · Cryptography & Network Security

RESEARCH EXPERIENCE

Student Researcher — Reinforcement Learning & Optimization, Presidency University *2025 – Present*

- Contributed to the implementation and evaluation of Proximal Policy Optimization (PPO) for navigation in a large 5,058 × 5,058 lunar-grid environment.
- Contributed to experimental analysis of reinforcement-learning performance and to a manuscript currently under review.

RESEARCH INTERESTS

Adaptive Computing · AI Systems · Reliable & Trustworthy AI · Intelligent Systems

SELECTED PROJECTS

Luna — AI-Powered Developer Assistant

Rust · LLM APIs · SQLite · Async Systems

- Built an AI-powered developer assistant in Rust combining conversational interaction, terminal execution, persistent memory, and multi-provider LLM integration.
- Designed a persistent SQLite-based memory system enabling retrieval of relevant context across sessions.
- Implemented safety mechanisms for potentially dangerous shell commands, requiring risk assessment and user confirmation before execution.
- Integrated multiple LLM providers while separating model interaction from core application logic, enabling providers to be switched without redesigning the system.

Sentinel — Software Supply-Chain Security Analyzer

Python · CycloneDX · OSV · CISA KEV · AST Analysis · LLMs

- Built a software supply-chain security analyzer for Python and Node.js projects that constructs dependency graphs, generates CycloneDX SBOMs, and identifies known vulnerabilities.
- Integrated OSV and CISA Known Exploited Vulnerabilities (KEV) data to identify vulnerable dependencies and known exploitation activity.
- Implemented AST-based reachability analysis to assess whether vulnerable Python dependencies are potentially used by application code.
- Developed a deterministic 0–100 supply-chain risk score based on vulnerability severity, exploitability, dependency exposure, and related security findings.
- Added grounded AI explanations and interactive queries over structured scan results, keeping security findings and scoring independent from LLM-generated interpretation.

Compass — Open-Source Issue Recommendation Platform

React · TypeScript · FastAPI · Python · GitHub API · Docker · LLMs

- Built and deployed a full-stack platform that recommends GitHub issues based on developer skills and experience, combining deterministic ranking with bounded LLM reasoning.
- Designed a deterministic confidence model using skill overlap, issue complexity, and repository health, with LLM reasoning limited to interpreting verified issue data.
- Implemented repository-structure verification to prevent LLM-generated explanations from referencing nonexistent files or paths.
- Developed a React/TypeScript frontend and FastAPI backend, with Docker-based development, GitHub Actions CI, and automated testing.

- Iterated on the architecture after identifying failure cases in LLM-generated repository paths, replacing model assumptions with verified repository data.

TECHNICAL SKILLS

Programming: Python · Rust · TypeScript · JavaScript · SQL

AI / Machine Learning: Machine Learning · Deep Learning · Reinforcement Learning · LLM APIs · RAG · LangGraph · Prompt Engineering

Software Development: React · FastAPI · REST APIs · SQLite · PostgreSQL · Git · GitHub

Systems / Infrastructure: Linux · Docker · GitHub Actions · Async Programming

Security / AI Systems: SBOM · Dependency Analysis · Static Analysis · OSV · CycloneDX